

Practical 11

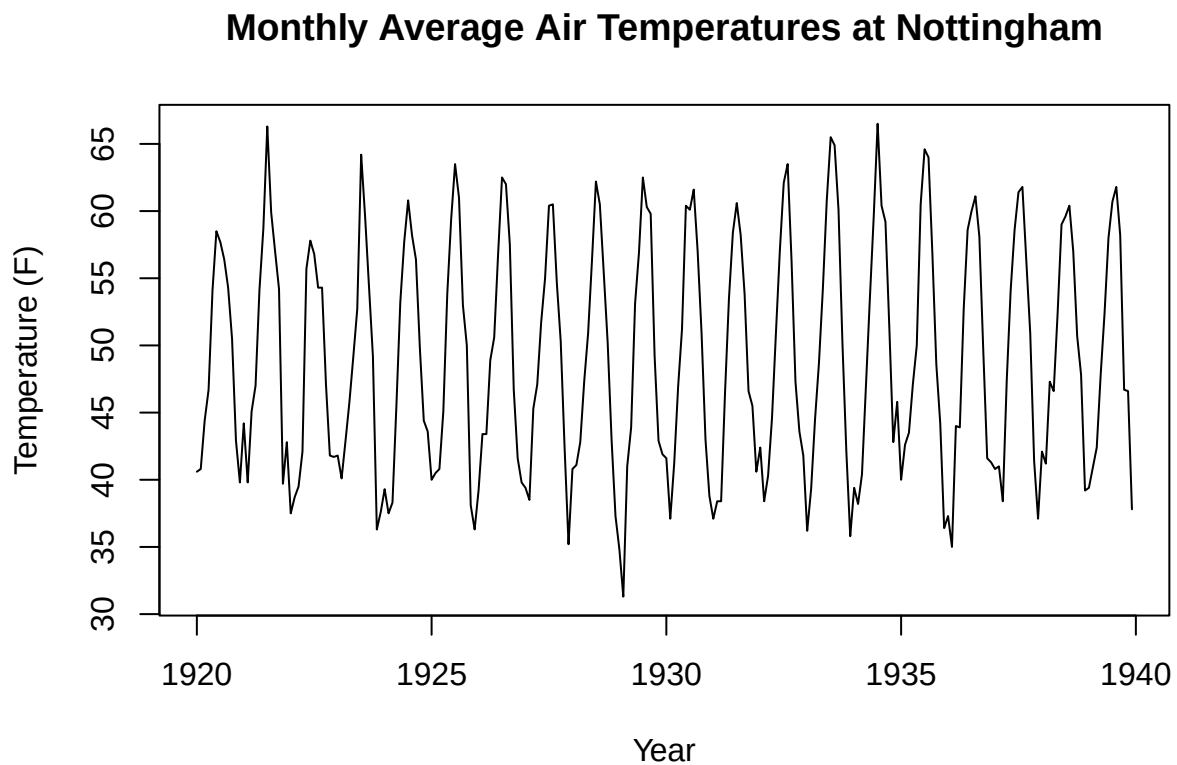
Rijul

2026-03-28

```
# Load dataset
data("nottem")

# Convert to time series object
ts_data <- nottem
```

```
# Plot the series
plot(ts_data, main="Monthly Average Air Temperatures at Nottingham",
      ylab="Temperature (F)", xlab="Year")
```



```
#Load forecasting libraries
library(forecast)
library(tseries)
```

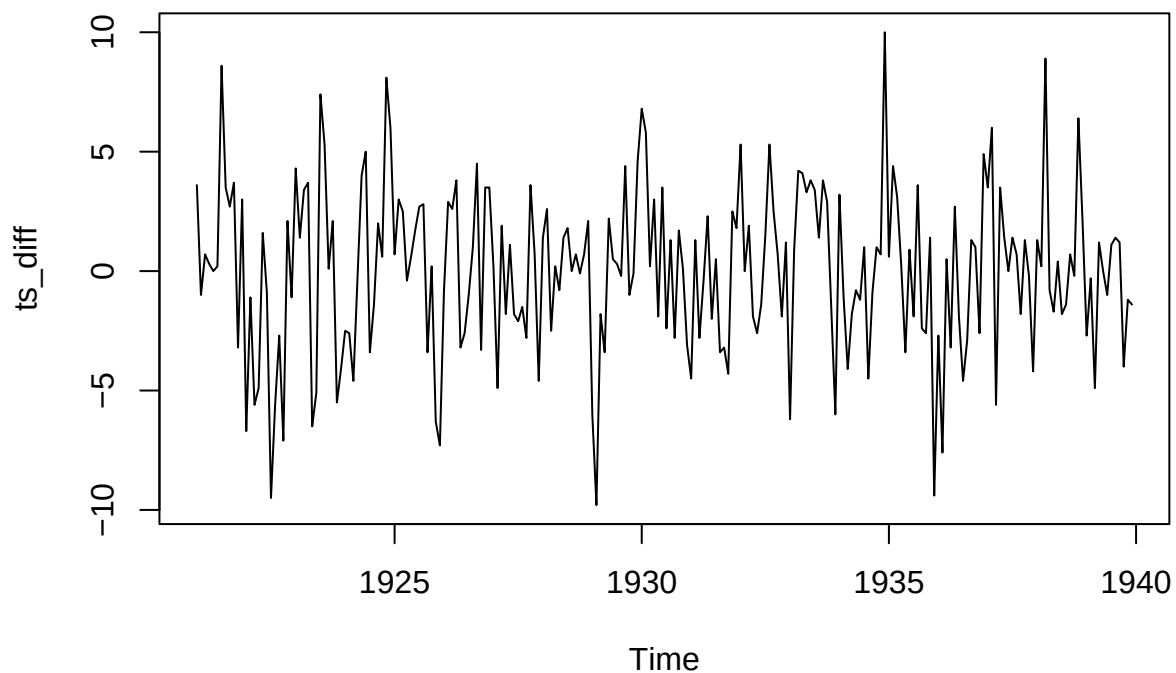
```
## Registered S3 method overwritten by 'quantmod':  
##   method      from  
##   as.zoo.data.frame zoo
```

```
#check for stationarity  
adf.test(ts_data)
```

```
## Warning in adf.test(ts_data): p-value smaller than printed p-value
```

```
##  
## Augmented Dickey-Fuller Test  
##  
## data: ts_data  
## Dickey-Fuller = -12.998, Lag order = 6, p-value = 0.01  
## alternative hypothesis: stationary
```

```
#apply seasonal differencing  
ts_diff <- diff(ts_data, lag=12)  
plot(ts_diff)
```



```
#Fit an ARIMA/SARIMA model  
model <- auto.arima(ts_data, seasonal=TRUE)  
summary(model)
```

```
## Series: ts_data
## ARIMA(1,0,2)(1,1,2)[12] with drift
##
## Coefficients:
##          ar1      ma1      ma2      sar1      sma1      sma2      drift
##          0.1562  0.0898  0.1113 -0.5324 -0.4925 -0.2391  0.004
## s.e.    0.3699  0.3651  0.1139  0.1911  0.2071  0.1880  0.004
##
## sigma^2 = 5.22: log likelihood = -516.48
## AIC=1048.96   AICc=1049.62   BIC=1076.4
##
## Training set error measures:
##              ME      RMSE      MAE      MPE      MAPE      MASE
## Training set -0.05647463 2.192395 1.695266 -0.3968809 3.647691 0.6295124
##              ACF1
## Training set 0.002370585
```

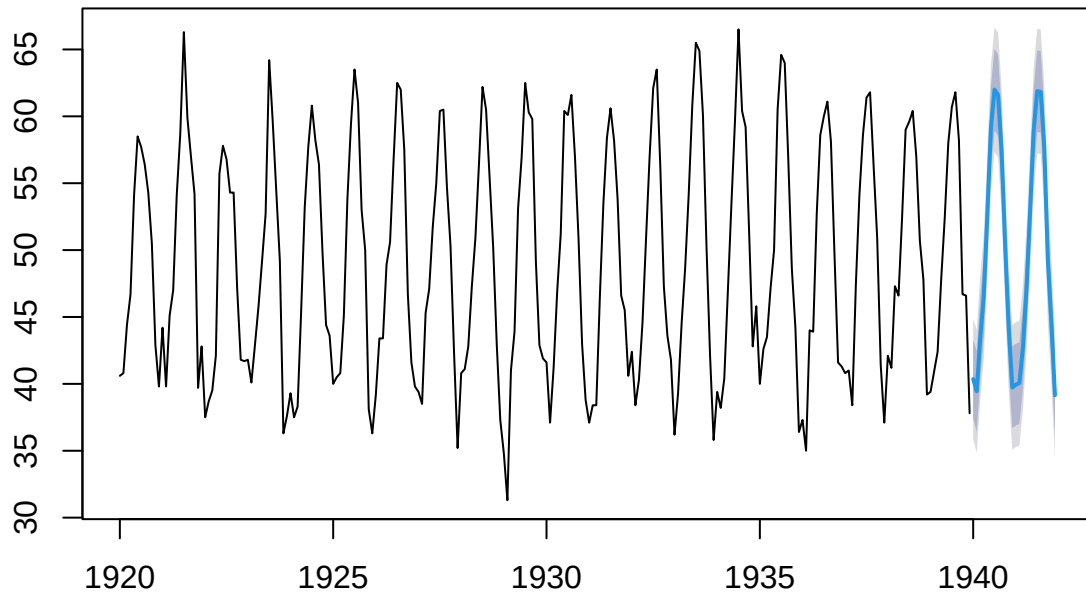
```
#forecast the values
forecast_values <- forecast(model, h=24)

print(forecast_values)
```

```
##      Point Forecast    Lo 80    Hi 80    Lo 95    Hi 95
## Jan 1940      40.35893 37.43091 43.28694 35.88091 44.83694
## Feb 1940      39.44870 36.43344 42.46396 34.83726 44.06014
## Mar 1940      43.45911 40.41218 46.50604 38.79923 48.11899
## Apr 1940      46.61631 43.56861 49.66401 41.95525 51.27737
## May 1940      52.81297 49.76525 55.86069 48.15189 57.47406
## Jun 1940      59.14794 56.10022 62.19566 54.48685 63.80903
## Jul 1940      61.99079 58.94307 65.03851 57.32970 66.65188
## Aug 1940      61.58756 58.53983 64.63528 56.92647 66.24864
## Sep 1940      57.38818 54.34045 60.43590 52.72709 62.04926
## Oct 1940      50.29230 47.24458 53.34002 45.63121 54.95339
## Nov 1940      44.15271 41.10499 47.20044 39.49163 48.81380
## Dec 1940      39.72837 36.68065 42.77609 35.06729 44.38946
## Jan 1941      39.94022 36.89165 42.98879 35.27784 44.60260
## Feb 1941      40.05869 37.01007 43.10731 35.39622 44.72115
## Mar 1941      42.73850 39.68986 45.78715 38.07601 47.40100
## Apr 1941      47.08000 44.03136 50.12864 42.41750 51.74249
## May 1941      52.81234 49.76370 55.86098 48.14984 57.47483
## Jun 1941      58.86558 55.81694 61.91422 54.20308 63.52807
## Jul 1941      61.87963 58.83099 64.92827 57.21714 66.54213
## Aug 1941      61.83258 58.78393 64.88122 57.17008 66.49507
## Sep 1941      57.68608 54.63743 60.73472 53.02358 62.34857
## Oct 1941      49.24880 46.20015 52.29744 44.58630 53.91129
## Nov 1941      44.56120 41.51256 47.60984 39.89871 49.22370
## Dec 1941      39.15417 36.10553 42.20281 34.49167 43.81666
```

```
plot(forecast_values)
```

Forecasts from ARIMA(1,0,2)(1,1,2)[12] with drift



```
#extract the forecasted values
forecast_values$mean
```

```
##           Jan      Feb      Mar      Apr      May      Jun      Jul      Aug
## 1940 40.35893 39.44870 43.45911 46.61631 52.81297 59.14794 61.99079 61.58756
## 1941 39.94022 40.05869 42.73850 47.08000 52.81234 58.86558 61.87963 61.83258
##           Sep      Oct      Nov      Dec
## 1940 57.38818 50.29230 44.15271 39.72837
## 1941 57.68608 49.24880 44.56120 39.15417
```